Executive Summary

One of the many goals in the UK's Industrial Strategy is to realise the productive potential of city regions across the nation. Glasgow City Region (GCR) has enormous economic potential with the presence of several industrial sectors, however, there are local issues which are constraining productivity and further growth.

Our analysis examined the local conditions associated with industry presence across approximately 350 local authorities in the UK. These national findings were then compared to GCR's current state. Nationally, the Professional and Business Services sector dominates with 46% of total IS8 employment. IS8 specialisation varies widely across UK city regions, with urban areas developing distinct sectoral strengths. Our analysis revealed that knowledge-intensive service sectors such as Creative Industries, Digital and Technology, and Professional Services tend to cluster together in the same locations, while manufacturing-oriented sectors concentrate in different types of areas (See A2 in Appendix). We also found that areas with dynamic business environments, high productivity, and strong health outcomes tend to have the greatest presence of industrial sector targets.

Between 2015 and 2024, GCR added approximately 37,710 IS8 jobs. Roughly two-thirds of that growth simply reflects national trends, and nearly all of the region's genuine local competitive advantage comes from Professional and Business Services alone. Glasgow's overall economic performance exceeds what its local conditions would predict, driven by the strength of its Financial and Professional Services sectors, as well as academic institutions. However, Creative Industries, one of the fastest growing sectors nationally, is Glasgow's largest area of underrealized productive potential.

Health is Glasgow's greatest barrier to growth. The region's under 75 mortality rate is nearly three times the national average. Closing the health gap could support roughly 50,000 additional high-growth sector jobs. Glasgow's university research base, its high concentration of fast-growing businesses, and its digital infrastructure provide strong foundations to build collaborative industrial interventions. However, addressing the health crisis is essential to improving the conditions for sector growth.

We recommend three connected interventions to resolve Glasgow's economic barriers through existing policy infrastructure. First, use existing innovation hubs like the Financial Regulation Innovation Lab and Glasgow City Innovation District to grow sectors where Glasgow is underperforming. Second, leverage Glasgow's universities to strengthen the talent pipeline into high-growth sectors. Third, extend Scotland's Fair Work First framework to tie innovation funding to support employment practices that raise livable wages and require higher working standards.

Introduction

The UK's Industrial Strategy identifies eight sectors (IS8) with the highest potential for productivity and growth. The key IS8 sectors include Advanced Manufacturing, Creative Industries, Professional and Business Services, Digital and Technology, Financial Services, Life Sciences, and Defence. This memo aims to understand which local conditions drive sector presence in the UK, and compares these results to the GCR to inform policy recommendations that incorporate the region's competitive advantages through existing infrastructure.

Data and Methodology

This analysis covers approximately 350 Local Authority Districts (LAD) across England, Scotland, and Wales. The Glasgow City Region comprises eight council areas: Glasgow City, East

Dunbartonshire, West Dunbartonshire, North Lanarkshire, South Lanarkshire, East Renfrewshire, Renfrewshire, and Inverclyde.

Our core datasets are employee counts from the Business Register and Employment Survey (2015-2024) and workplace counts from UK Business Counts (2016-2025), both mapped to IS8 sectors using the government's official sector definitions. These were complemented by approximately 55 local indicators from the Office of National Statistics (ONS) covering economic productivity, health, education, and digital infrastructure, as well as Higher Education Statistics Agency (HESA) data on university research income and student counts.

In our analysis, employment and business shares are used as proxies for IS8 sector presence and growth, measuring the proportion of each IS8 sector in a local area's economy. Nationally, we mapped IS8 employment and business concentration across UK local authorities, grouped areas with similar industrial profiles using cluster analysis, and identified which sectors tend to co-locate. We then used a predictive model trained on local indicators to identify the conditions most strongly associated with IS8 presence and growth.

Locally, we applied the national model to Glasgow to assess where the region over- and underperforms relative to what its local conditions would predict, and quantified the job impacts of addressing Glasgow's key barriers or losing its key strengths. Full methodological details, model specifications, and supporting figures are provided in the technical appendix.

National Analysis

Professional and Business Services dominates UK IS8 employment, comprising 46% of the total in 2024, followed by Digital and Technology (18%), Financial Services (14%), Creative Industries (14%), and Advanced Manufacturing (7%). Life Sciences and Defence together account for just over 1%. With England's larger population, it tends to dominate total IS8 employment.

Among the seven regions studied, West of England specialises in Defence, West Midlands in Advanced Manufacturing, South Yorkshire in Life Sciences, and Greater Manchester in Professional and Business Services. Glasgow City Region, Cardiff-Newport, and West Yorkshire each have their highest concentration in Financial Services (see Figure 3).

To identify which local authorities share similar industrial characteristics, we grouped all UK local authorities based on their sector concentrations (see Figure 5). The largest group comprises 249 smaller and more rural local authorities with limited IS8 presence. A second group of 58 local authorities represents knowledge-intensive service hubs with strong concentrations of Creative, Digital, and Professional Service industries. Major cities including London, Manchester, Leeds, and Glasgow City all fall into this category, indicating that Glasgow City shares a similar industrial profile to other leading urban economies in the UK. Six of the eight GCR council areas fall into the rural/agricultural group, highlighting the sharp divide between Glasgow City's knowledge economy and the wider region's industry.

Our analysis also identified which IS8 sectors tend to appear together in the same locations (see Figure 6). The strongest relationship was between Creative Industries and Digital and Technology, followed by Digital and Technology with Professional and Business Services. By contrast, Advanced Manufacturing tends to concentrate in different types of areas than knowledge-intensive service sectors. These co-location patterns suggest that investing in one of these linked sectors can create conditions favourable for simultaneous growth.

A predictive model was also used to identify the local indicators associated with IS8 presence and growth in the UK such as health, education, and economic factors. The top three

predictive indicators for sector employment share were high growth enterprises, under 75 mortality rate, and Gross Value Added (GVA) per hour. High growth enterprises and GVA per hour both positively impact the presence of IS8 sectors, while under 75 mortality rate is a top indicator that negatively impacts the presence of IS8 sectors.

The top three local indicators identified for predicting business share were level 3+ qualifications, high growth enterprises, and apprenticeship achievements. While level 3+ qualifications and high growth enterprises positively impact the presence of IS8 sectors, the apprenticeship achievements indicator appears to have an unusual negative impact on the presence of IS8 Business Share. Our models indicate that high growth and productivity are associated with higher concentrations of IS8 sectors, whereas low health outcomes predict lower concentrations of these industries.

Additionally, local indicators data was used to predict the employment share within each IS8 sector (see Figure 8). Creative Industries, Digital and Technology, Financial Services and Professional and Business Services all share a similar set of predictors such as high growth enterprises and GVA per hour. This suggests that these sectors have higher presence in growing areas. Research funding for universities consistently appeared as a top predictor for multiple sectors including Creative Industries, Financial Services, and Digital and Technology. The findings' main takeaway suggests that high growth and education are top predictors for IS8 sector concentration.

Glasgow City Region Analysis

To target IS8 sector presence and growth for the GCR, deeper analysis was necessary to identify region-specific indicators.

Our analyses confirmed known health issues. Glasgow's health outcomes are significantly below the national average, with an under 75 mortality rate nearly three times the UK average and male healthy life expectancy roughly six years below the national mean. However, GCR benefits from above-average scores in education factors, including university research income in the 97th percentile nationally. This positions Glasgow uniquely amongst UK city regions, combining severe health disadvantages with strong educational foundations.

When using the national model to predict Glasgow's current economic state, the region's real IS8 employment share is 4.2 percentage points higher than its predicted value (see Figure 9). This overperformance is driven by Financial Services and Professional and Business Services. The Financial Services sector had the largest gap between its real and predicted values, with a real value 3.4 percentage points above prediction. This likely reflects the city's above-average concentration in Financial Services, anchored by the International Financial Services District (IFSD) which hosts major institutions including JP Morgan and Barclays (Glasgow City Council, 2024).

Despite Glasgow's overall real employment share outperforming its predicted value, there are specific sectors where the region is performing below its predicted value. Creative Industries, nationally the fastest growing IS8 sector, represents the largest shortfall in Glasgow with a real value that is 1.1 percentage points below prediction. This gap represents significant unrealised potential. Life Sciences also falls below expectation, despite Renfrewshire holding over three times the national average concentration in that sector (see Figure 12). This suggests that regional assets exist but are not yet translating into broader employment gains.

To understand the dynamics of GCR's IS8 growth, we conducted a shift-share decomposition, breaking down employment change into national growth, industry mix, and local competitiveness (see Figure 10). National growth captures how much of Glasgow's change is driven

by overall UK trends. Industry mix captures the effect of Glasgow's particular sector composition. Local competitiveness isolates the region's unique advantages, such as its specialised workforce, institutional anchors, and local policy environment, that give it an edge in certain industries.

From 2015 to 2024, GCR's IS8 employment grew by approximately 37,710 jobs. However, roughly 65% of that growth reflects the national expansion of IS8 sectors, suggesting that much of Glasgow's growth can be attributed to riding national tailwinds. A further 20% comes from having a favourable industry mix, being concentrated in sectors that are also growing nationally. Only around 15% of the growth can be attributed to genuine local competitiveness.

Professional and Business Services accounts for 68% of total IS8 growth and nearly all of the positive local competitiveness effect. By contrast, Financial Services, one of GCR's stronghold sectors, lost approximately 1,400 jobs over the period despite positive national trends, pointing to a negative local competitiveness effect. Glasgow is also Scotland's largest creative economy, though as our analysis reveals, this sector is underperforming relative to what local conditions would predict (Invest Glasgow) due to a negative score in local competitiveness. Digital and Technology, however, is a bright spot, growing across all three components (see Figure 10). Glasgow's local competitive advantages are concentrated almost entirely in one sector, and broadening that competitiveness beyond Professional and Business Services is essential to sustaining long-term growth.

Strengths and Barriers

A deeper analysis of GCR further informs understanding which characteristics are barriers that hinder growth, and which are regional strengths valuable to promoting growth (see Figure 11). Glasgow's largest strength is its research funding, stemming from its rich university presence. Level 3+ qualifications also showed itself to be a strength in our analysis, further emphasizing university presence as a driver for economic growth. Universities act as anchor institutions, enabling economic growth through investment, research, and knowledge transfers (McCann et al., 2023). It is unsurprising that Glasgow's universities have such a sizable impact. The second greatest regional strength is the number of high growth enterprises, which is almost four times higher than the UK average. Another strength in the region is broadband availability, highlighting Glasgow's Digital and Technology sector and general connectivity strengths.

Most of Glasgow's barriers to economic growth are related to poor health outcomes, with results showing that the under 75 mortality rate in the region is almost three times higher than the UK average (see Figure 11). Other health barriers that proved themselves important in our analysis are smoking prevalence and male healthy life expectancy. The data reinforced a concept already known as the Glasgow Effect, a term used to refer to the poor health standards and low life expectancy in Glasgow (Landy et al, 2010).

Furthermore, we quantified these strengths and barriers as the number of jobs that could be lost or gained. If all Glasgow's strengths were to be lost, the region would lose more than 160,000 jobs. However, by addressing the barriers, the region could gain more than 60,000 additional jobs, with health related barriers alone accounting for over 50,000 of those openings.

Policy Recommendations

Our analysis points to three mutually reinforcing policy interventions, each grounded in the findings above and deliverable through existing infrastructure in Glasgow.

Target sector adjacencies through existing innovation hubs. Financial Services, a sector that distinguishes Glasgow nationally and is relatively concentrated in the region, is declining. Our co-location analysis identified strong sector adjacencies that Glasgow can exploit. We recommend

extending funding for the existing fintech accelerator through the Financial Regulation Innovation Lab (FRIL). Given that FRIL has already attracted investment and demonstrated strong returns to previous funding, we recommend using it to channel Glasgow's traditional financial services anchor toward Digital and Technology growth. For Creative Industries, Glasgow's greatest sector of unrealised potential, we recommend expanding the Creative-Digital crossover programme through the Glasgow City Innovation District (GCID). GCID already hosts over 1,600 organisations and is well positioned to support firms operating at the intersection of these two highly correlated sectors. Together, these interventions target the specific sector adjacencies identified in our national analysis.

Utilise Glasgow's university presence as connective tissue. Research funding emerged as one of the strongest national predictors of IS8 presence. The University of Glasgow and the University of Strathclyde together generate £131 million in research funding (HESA Research Income Data, 2024). However, GCR reports low graduate retention rates in key fields such as engineering and business (Glasgow City Region, 2025). Formalising industry partnerships through the innovation hubs identified above can help train and retain a qualified IS8 workforce. Glasgow's universities do not just supply workers; they generate the applied research environment that attracts IS8 firms.

Address health barriers through an economic framework. Health outcomes are the single largest negative predictor of IS8 presence in Glasgow. Our counterfactual analysis estimates that addressing health barriers could support approximately 50,000 additional IS8 jobs. Rather than treating health as a standalone social issue, we propose extending the Scottish Government's Fair Work First framework, which makes public sector grants conditional on living wage and fairness requirements (Scottish Government, 2023), to Innovation Accelerator, Investment Zone, and City Deal funding distributed through innovation hubs. This approach aims to address Glasgow's most significant economic barrier by raising living standards to improve local productivity (McCann et al, 2022).

All three pillars flow through existing infrastructure and reinforce each other. Better health expands the available workforce. A stronger workforce attracts more IS8 firms. More IS8 firms generate the research partnerships that universities need to sustain their funding advantage. Glasgow does not need new institutions, it needs coordination.

Technical Appendix

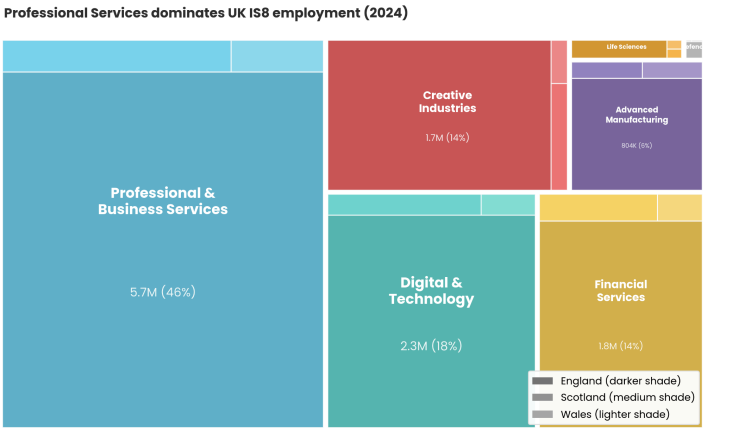
A1. Data and Variable Construction

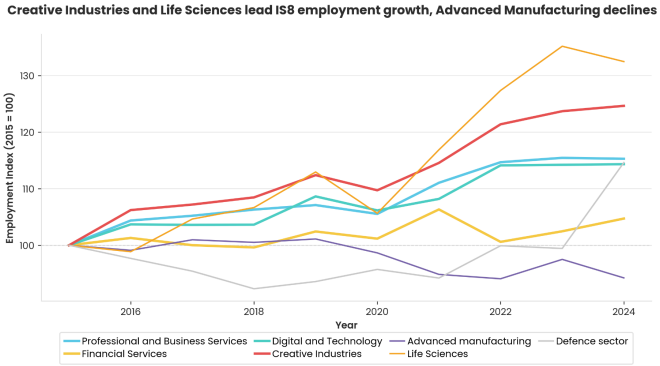
Brief description of the datasets, how they were merged, and key decisions made:

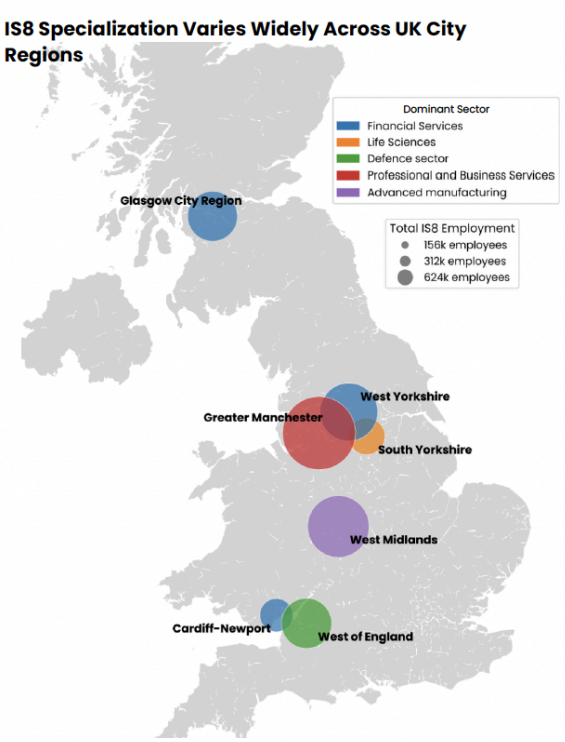
- BRES employee counts (2015-2024) and UK Business Counts (2016-2025) at the LAD level, mapped onto IS8 sectors via the official SIC lookup
- ONS local indicators (~55 variables, ~35 England-only), supplemented with HESA research funding data
- Missingness and variable exclusions: Nine columns with zero non-null values were dropped entirely including Northern Ireland-only duplicates and non-LAD variables. Of the remaining numeric indicators, 22 were completely missing for all Scottish LADs, including England specific education metrics (KS2 attainment, GCSE). We sourced Scottish equivalents like median house prices, under 75 mortality, and health life expectancy.
- Location quotient construction: the BRES employee counts data arrives at the individual SIC code level. We first aggregated employment from SIC codes to the IS8 sector level for each LAD using the official IS8 SIC Lookup, then computed location quotients as: LQ= (local sector employment / local total employment) / (national sector employment / national total employment), where total employment refers to all-economy employment across all SIC sectors, not just IS8 sectors. National totals were constructed by summing across all LADs (excluding country level aggregates). An LQ above 1.0 indicates greater concentration than the national average.

A2. National Descriptive Analysis

Figure 1 (top left): UK IS8 Employment Composition; Figure 2 (bottom left): UK IS8 sector growth from 2015 to 2024 (2015 = 100); Figure 3 (right): City region specialisation map showing dominant IS8 sector by region and total IS8 employment size







A3. K-Means Clustering

- Input features: location quotients across IS8 sectors at the LAD level, standardised before clustering.
- Selecting k: tested k=2 through 10. Reported elbow plot (inertia vs k) and silhouette scores. K=5 was selected based on the elbow point and one of the highest silhouette scores among the interpretable solutions.

Figure 4: Elbow plot and silhouette score plot

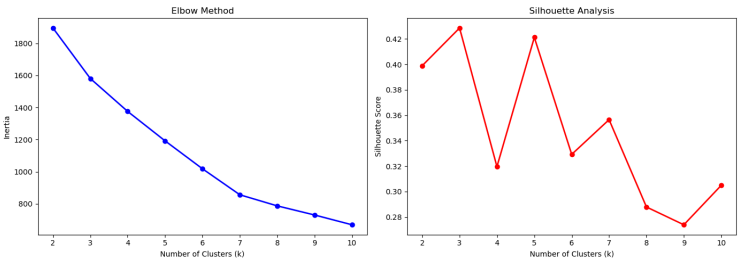
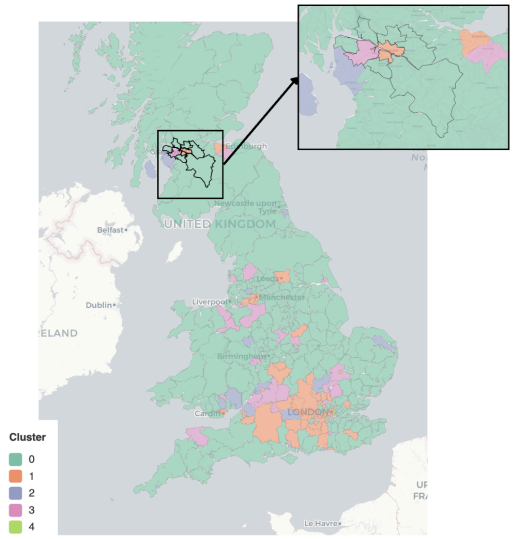


Figure 5: Cluster profiles showing cluster number, count of LADs, and description based on LQ concentration

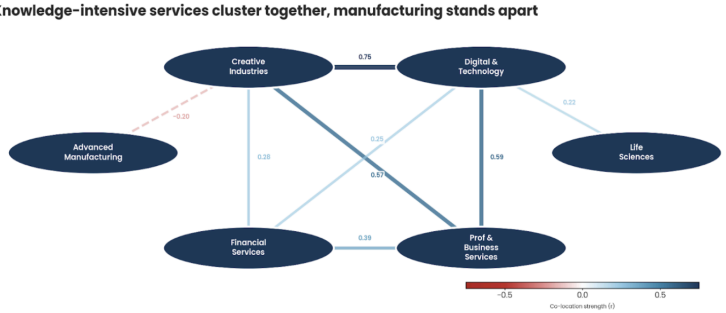


Cluster	LADs	Description
0	249	Smaller towns, rural/agricultural economies
1	58	Knowledge-intense service hubs: creative, digital, and professional service industries
2	13	Defence and Manufacturing strongholds
3	28	Life Sciences and Manufacturing Specialists
4	2	Financial Services in London

A4. Inter-Industry Co-Location Analysis

- Pairwise Pearson correlations of IS8 sector LQs across all LADs. Values range from -1 to +1 where positive values indicate sectors that tend to concentrate in the same places and negative values indicate sectors that are located in different types of areas.
- Key finding: knowledge-intensive services cluster tightly (Creative-Digital correlation of 0.75, Professional Services-Digital at 0.59), while Advanced Manufacturing is negatively correlated with Creative Industries (-0.2) and weakly linked to the service sectors.

Figure 6: Co-location strength between sectors



A5. Elastic Net Regression

- Model specification: dependent variables were IS8 employment share and business share (total and by sector). Independent variables were standardized local indicators.

- Given the low R^2 values for employment and business growth, we will be using employee and business shares as a proxy for growth and our main outcome variables throughout the analysis

- Why elastic net over LASSO: Many indicators are correlated (GVA per hour, weekly pay, GDHI all capture economic prosperity). Elastic net combines L1 and L2 penalties, handling correlated predictors better than pure LASSO by sharing coefficients rather than arbitrarily dropping one. We ran both and compared results for robustness.
- Cross validation and tuning: We used ElasticNetCV with a 10-fold cross-validation over five L1 ratios (0.1, 0.3, 0.5, 0.7, 0.9) and also used a 70/30 train-test split.

Two model specifications

- Model A (UK-wide): ~350 LADs, 25 independent variables available across England, Scotland, and Wales (including GVA per hour, enterprise metrics, broadband availability, health indicators, wellbeing measures, and HESA variables)
- Model B (England-only): ~296 LADs, 36 independent variables including 13 additional England-only indicators (transport accessibility, early years and GCSE education metrics, obesity measures, cancer diagnosis, net housing additions, and FE participation)

Table 1: R^2 (train and test) for LASSO and Elastic Net across both Model A and Model B for each outcome variable

	Model A (UK)				Model B (England-only)			
Outcome	Lasso Train	Lasso Test	ENet Train	ENet Test	Lasso Train	Lasso Test	ENet Train	ENet Test
Total IS8 Employment Share	0.7	0.545	0.704	0.62	0.62	0.37	0.616	0.373
Total IS8 Employment Growth	0.174	0.179	0.177	0.23	0.215	0.045	0.227	0.046
Total IS8 Business Growth	0.72	-0.274	0.558	-0.027	0.397	0.077	0.303	0.248
Total IS8 Business Share	0.913	0.754	0.913	0.757	0.883	0.808	0.881	0.828
Advanced Manufacturing Emp Share	0.25	0.321	0.222	0.372	0.408	-0.387	0.374	-0.133
Creative Industries Emp Share	0.748	0.603	0.726	0.604	0.7	0.411	0.699	0.415
Defence Sector Emp Share	0	-0.13	0	-0.13	0	-0.054	0	-0.054
Digital and Technology Emp Share	0.585	0.614	0.619	0.568	0.586	0.099	0.568	0.198
Financial Services Emp Share	0.219	0.468	0.239	0.564	0.072	0.156	0.075	0.161
Life Sciences Emp Share	0.293	0.1	0.289	0.105	0.079	-0.014	0.084	-0.01
Professional and Business Services Emp Share	0.472	0.492	0.484	0.543	0.531	0.142	0.535	0.139

Figure 7: Predictors for elastic net model of IS8 employment and business share

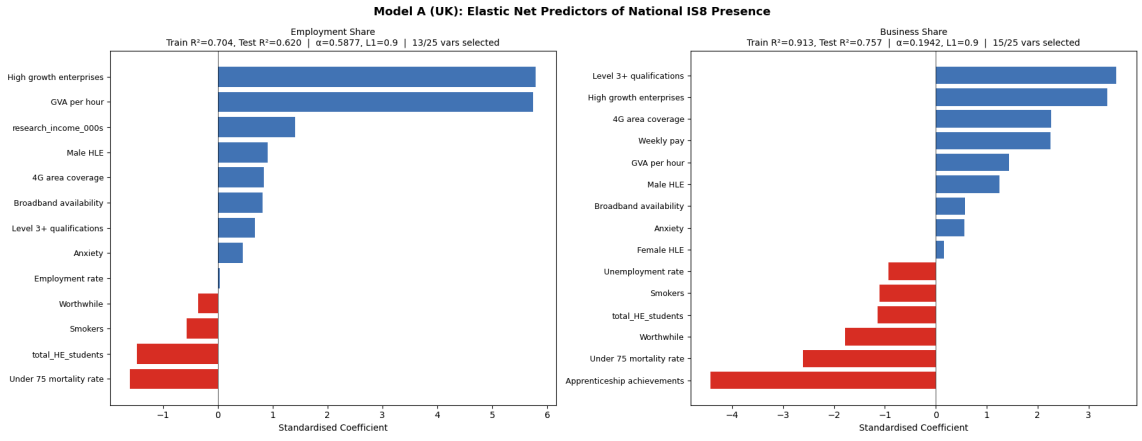
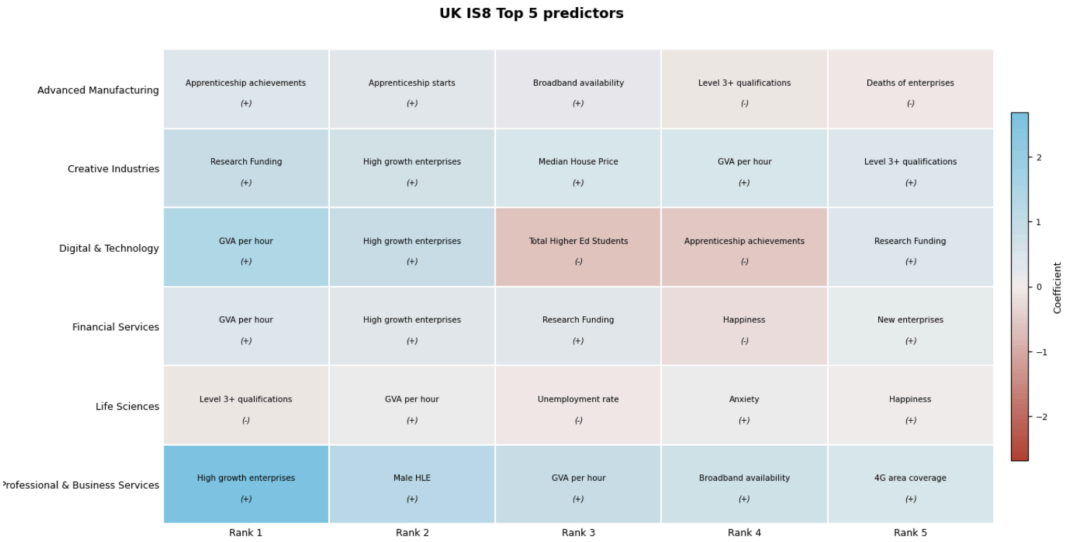


Figure 8: Elastic net top 5 predictors by IS8 sector, (+) positive association, (–) negative association

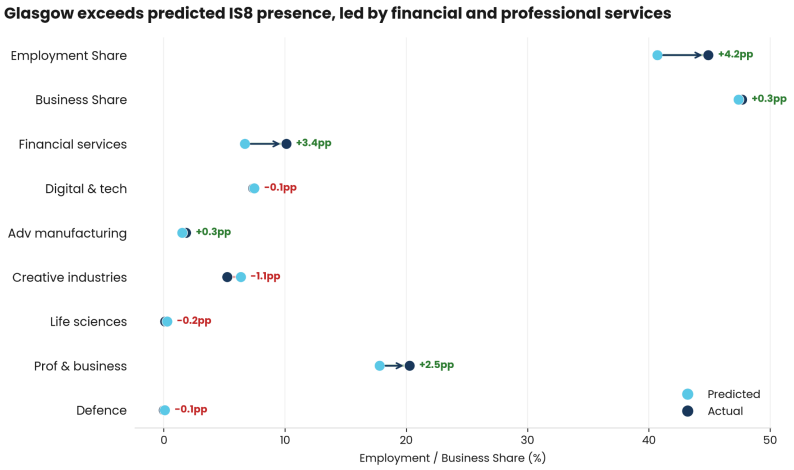


Note: Defence was not included in this table due to its -R^2 values and low sample size. The indicator "Apprenticeship achievements" includes data from England only.

A6. Glasgow's Predicted vs Actual Performance

Applied Model A elastic net coefficients to Glasgow's local indicator values from the most recent year available to generate predicted IS8 employment and business shares. The residual (actual minus predicted) indicates over or underperformance relative to what local conditions would suggest.

Figure 9: Glasgow's Actual vs Predicted Values



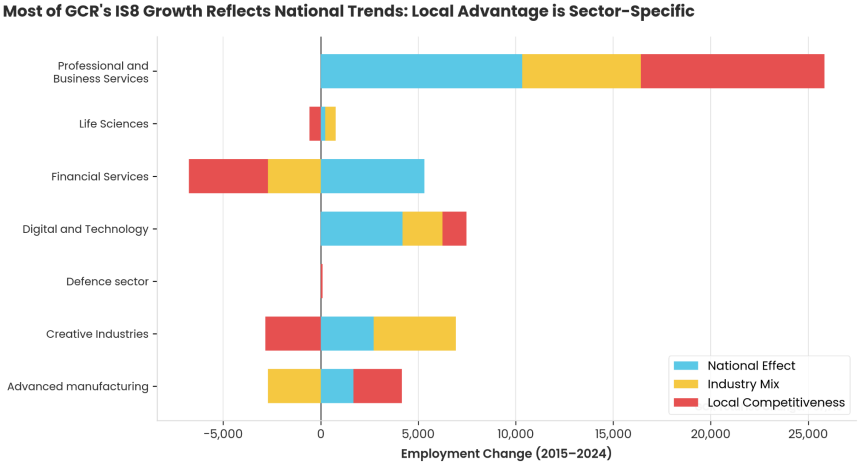
A7. Shift-Share Decomposition

Method: Three-component shift-share analysis comparing GCR IS8 employment change (2015-2024) against the national benchmark.

Formula:

- National effect = base employment x national growth rate
- Industry mix = base employment x (sector national growth rate - overall national growth rate)

Figure 10: Shift-share analysis by IS8 sector

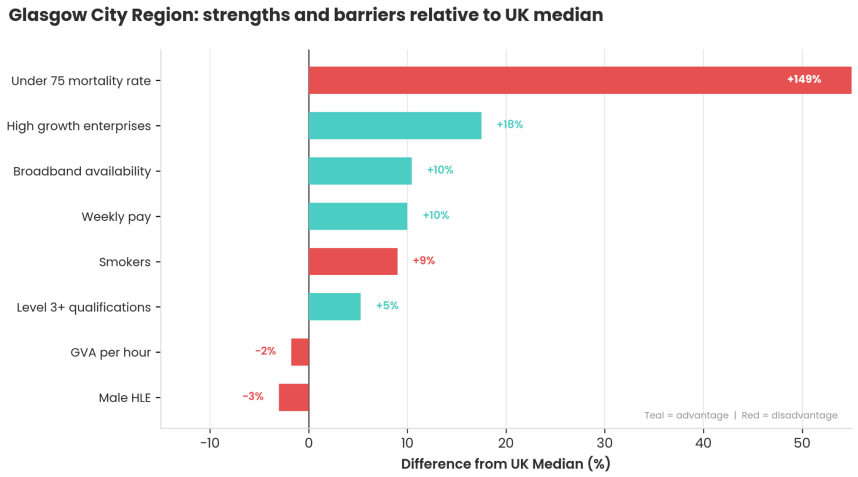


- Local competitiveness = base employment x (local sector growth rate - sector national growth rate)

A8. Counterfactual Analysis

- Method: for each local indicator where Glasgow differs from the UK average, we substituted the UK average value into the elastic net model and computed the change in predicted IS8 employment share. This was converted to estimated jobs using GCR's total employment base of 883,750.
- Limitations: this is a partial equilibrium which assumes that indicators can be changed independently, that cross-sectional relationships apply to within-area changes over time, and that the elastic net coefficients are stable. These are strong assumptions and the job estimates should be interpreted as illustrative of relative magnitudes rather than precise forecasts.

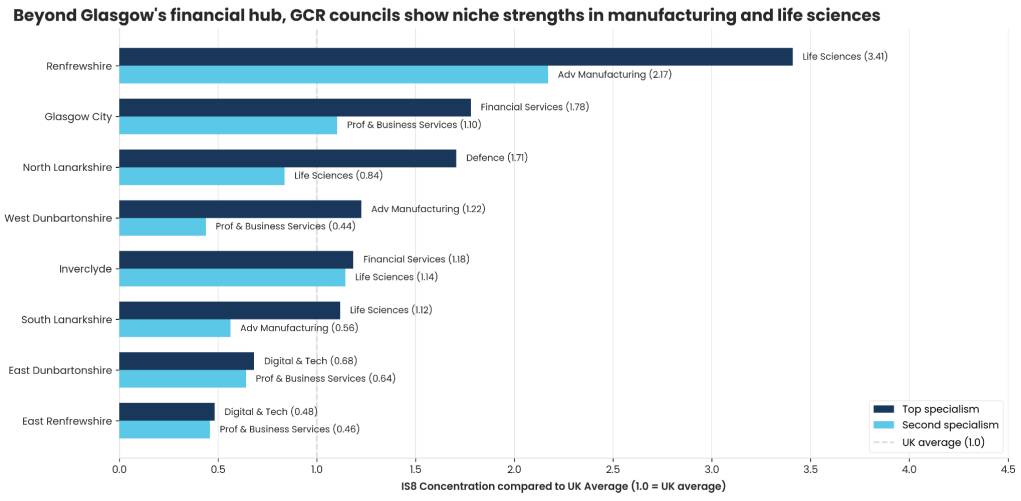
Figure 11: Individual barrier and strength impacts in estimated jobs



A9. Within-GCR Analysis

- Location quotients for each of the 8 GCR council areas, showing the concentration of IS8 sectors relative to the UK average. Glasgow City dominates with 68% of GCR's IS8 employment, while surrounding councils show distinct specialisms.

Figure 12: Location quotients by council area with the top and second specialisms marked



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